

Civil Engineering

Trigonometry and Complex Numbers

Revision Questions

Use these questions for CAT and summative assessment preparation.

Formula Practice

1. Apply the formula below in a technical example.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

2. Apply the formula below in a technical example.

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

3. Apply the formula below in a technical example.

$$c^2 = a^2 + b^2 - 2ab \cos C$$

4. Apply the formula below in a technical example.

$$z = a + bi$$

Applied Practice

5. Compute the height of a structure from angle of elevation.
6. Solve a triangular road reserve survey problem.
7. Convert a complex number to polar form.